Final Exam
FI542E: Exchange Markets

	First name	Family name
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Answer:

1. Data Description

We pull monthly data of FRED for Mexico and France, from January 2020 until July 2024. After align the three series, we end with 55 observations usable, and the endpoint is not a deliberate election, some series of FRED have lag in the publication, so July 2024 was just the last month with complete data in all the variables when we run the download in early 2026.

The variables are the Consumer Price Index for Mexico (MEXCPIALLMINMEI) and France (FRACPIALLMINMEI), plus the nominal exchange rate MXN/EUR, FRED do not publish MXN/EUR directly, so we construct it multiplying the daily MXN/USD rate (DEXMXUS) by the daily USD/EUR rate (DEXUSEU), then we average each daily series to monthly to match CPI frequency and the explanatory variable of the regression is the ratio of CPI, considering Mexican CPI divided by French CPI.

2. Model

The theoretical start point is the absolute purchasing power parity and, he main claim is that the nominal exchange rate between two countries should in equilibrium be equal to the ratio of the price levels, considering that if Mexico price level is 20% more high than France, the rate peso/euro should reflect that difference and no more, no less.

$$MXN/EUR = a \times (CPI_{mx} / CPI_{fr}) + b$$

The strict absolute PPP impose two conditions in this regression: the slope coefficient (a) should be equal to 1, and the intercept (b) should be equal to 0 so the more relaxed prediction is just that the slope should be positive and rising relative prices in Mexico should weaken the peso and push the MXN/EUR quote higher and we estimate the equation using OLS by the excel data analysis

3. Estimation Results

Table 1: Regression Statistics

Statistic	Value	Statistic	Value
Multiple R	0.8110	Observations	55
R^2	0.6577	F-statistic	101.8301
Adjusted R^2	0.6512	Significance F	0.0000
Standard Error	1.5407		

Table 2: ANOVA

Source	df	SS	MS	F
Regression	1	241.7198	241.7198	101.8301
Residual	53	125.8091	2.3738	
Total	54	367.5289		

Table 3: Coefficients

Variable	Coeff.	Std. Error	t Stat	p-value	Lower 95%	Upper 95%
Intercept	89.0803	6.6731	13.3493	0.0000	75.6958	102.4647
CPI Ratio	-54.7220	5.4228	-10.0911	0.0000	-65.5988	-43.8452

4. Interpretation

On its surface the model looks reasonably good fitted. R^2 of 0.66, f-statistic of 101.8 ($p < 0.001$), both coefficients statistically significant at any conventional threshold and if we stopped reading here we might think something was normal but the problem is the sign.

PPP predict a positive slope and well when the Mexican prices rise more fast than the French prices, the peso should depreciate itself so more pesos per euro and also that came out of the regression was -54.72. Over the 2020-2024 window, the CPI ratio climbed steadily (Mexican inflation consistently outpaced French inflation), but the peso in reality strengthen against the euro and the regression picked up exactly this pattern and returned a negative slope, which invert directly what the theory predicts

The magnitude makes things worse because the absolute PPP require a slope of 1 and intercept of 0 then our intercept sits at 89.08, with a t-statistic of 13.35 so there is a large baseline component in the MXN/EUR rate that the CPI ratio do not explain nothing about and here the slope being negative and large in absolute value simply rules out any version of the theory.

Absolute PPP does not hold for Mexico and France during this period.

5. Factors Explaining the Deviation

- First, interest rates, Mexico increase a lot the rates after COVID, above 11% at some point, while Europe kept rates very low until 2022 and that gap made it attractive to put money in Mexico, which helped the peso stay strong even with higher inflation. This kind of carry trade can keep a currency overvalued for years.
- Second, remittances, Mexico got a lot of money from workers abroad, around $64 billion in 2024. That money come every month no matter what the prices are doing, and this supports the peso in a way that PPP just doesn't account for.
- Third, nearshoring, many companies moved production to Mexico after 2020, that brought more foreign investment and more demand for pesos, again, this has nothing to do with inflation.
- There's also a problem with the data itself because CPI includes things like rent and healthcare that don't trade across borders so then if prices for those go up, the exchange rate doesn't have to move so if we see it like that using CPI to test PPP isn't perfect.
- Finally, 55 months is a short period, and the important thing here was a very unusual one. COVID, global inflation, fast rate hikes and ideally PPP usually needs many years of data to show up clearly and in this sample, there was too many things happening.

Appendix 1 (References)

Banco de México. (n.d.). *Summary table details (SIE, Banco de México).* Sistema de Información Económica.

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Banco de México. (2025, December 18). *Minutes number 121.* https://www.banxico.org.mx/publications-and-press/minutes-of-the-board-of-governors-meetings-regardi/%7B354E7560-2CD2-AF3C-5E69-4BE4448D8012%7D.pdf

European Central Bank. (n.d.). *Key ECB interest rates.* https://www.ecb.europa.eu/stats/policy_and_exchange_rates/key_ecb_interest_rates/html/index.en.html

International Monetary Fund. (2024, October 30). Statement by Mr. Guerra, Executive Director for Mexico and Ms. Posadas Bolanos, Advisor to Executive Director. https://www.elibrary.imf.org/downloadpdf/view/journals/002/2024/317/article-A003-en.pdf

Appendix 2 (Database + Code)

In [ ]:
```
# !pip install pandas
# !pip install pandas_datareader

import pandas as pd
from pandas_datareader import data as web

start = "2019-01-01"
end = "2026-02-28"


# FRED series
series = {
    "cpi_mex": "MEXCPIALLMINMEI",
    "cpi_fra": "FRACPIALLMINMEI",
    "rate_mex": "IRSTCI01MXM156N",   # monthly Mexico interbank/call money proxy
    "rate_ecb": "ECBDFR",            # daily ECB deposit facility rate
    "mxn_usd": "DEXMXUS",            # daily MXN per USD
    "usd_eur": "DEXUSEU"             # daily USD per EUR
}

# Download
raw = pd.concat(
    [web.DataReader(code, "fred", start, end).rename(columns={code: name})
     for name, code in series.items()],
    axis=1
)
```

In [ ]:
```
# Convert daily series to monthly
monthly = pd.DataFrame(index=pd.date_range(start=start, end=end, freq="MS"))

# Monthly series already in FRED
monthly["cpi_mex"] = raw["cpi_mex"]
monthly["cpi_fra"] = raw["cpi_fra"]
monthly["rate_mex"] = raw["rate_mex"]

# Daily to monthly average
monthly["rate_ecb"] = raw["rate_ecb"].resample("MS").mean()
monthly["mxn_usd"] = raw["mxn_usd"].resample("MS").mean()
monthly["usd_eur"] = raw["usd_eur"].resample("MS").mean()

# Build MXN per EUR
monthly["mxn_eur"] = monthly["mxn_usd"] * monthly["usd_eur"]

# Optional: monthly inflation rate (% change from same month previous year)
monthly["infl_yoy_mex"] = monthly["cpi_mex"].pct_change(12) * 100
monthly["infl_yoy_fra"] = monthly["cpi_fra"].pct_change(12) * 100
```

```
/var/folders/0w/gy1wlvdn00sd9463511ystdc0000gn/T/ipykernel_42866/844076801.py:18: FutureWarning: The default fill_method='pad' in Series.pct_change is deprecated and will be removed in a future version. Either fill in any non-leading NA values prior to calling pct_change or specify 'fill_method=None' to not fill NA values.
  monthly["infl_yoy_mex"] = monthly["cpi_mex"].pct_change(12) * 100
/var/folders/0w/gy1wlvdn00sd9463511ystdc0000gn/T/ipykernel_42866/844076801.py:19: FutureWarning: The default fill_method='pad' in Series.pct_change is deprecated and will be removed in a future version. Either fill in any non-leading NA values prior to calling pct_change or specify 'fill_method=None' to not fill NA values.
  monthly["infl_yoy_fra"] = monthly["cpi_fra"].pct_change(12) * 100
```

In [ ]:
```
# Keep final project dataset
final_df = monthly[[
    "cpi_mex", "cpi_fra",
    "infl_yoy_mex", "infl_yoy_fra",
    "rate_mex", "rate_ecb",
    "mxn_eur"
]].copy()

# Optional but recommended
final_df = final_df.dropna()

# Preview
print(final_df.tail())

# Save file
import os

path = os.path.join(os.path.expanduser("~"), "Downloads", "mexico_france_macro_monthly.csv")
final_df.to_csv(path, index=True)

print("Saved at:", path)
```

```
             cpi_mex  cpi_fra  infl_yoy_mex  infl_yoy_fra  rate_mex  rate_ecb  \
2024-03-01  152.9470   119.47      4.420997      2.294717      8.79  4.000000
2024-04-01  153.2561   120.07      4.653176      2.187234      8.64  4.000000
2024-05-01  152.9721   120.11      4.686786      2.273501      8.63  4.000000
2024-06-01  153.5505   120.20      4.976075      2.167446      8.64  3.841667
2024-07-01  155.1579   120.42      5.566181      2.302268      8.57  3.750000

              mxn_eur
2024-03-01  18.228517
2024-04-01  18.017035
2024-05-01  18.164572
2024-06-01  19.583366
2024-07-01  19.632517
Saved at: /Users/inigoocampodevecchi/Downloads/mexico_france_macro_monthly.csv
```

In [ ]:
```
final_df.shape
```

Out[ ]:
```
(55, 7)
```

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.8110
R Square	0.6577
Adjusted R Sc	0.6512
Standard Erro	1.5407
Observations	55.0000

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	241.7198	241.7198	101.8301	0.0000
Residual	53	125.8091	2.3738		
Total	54	367.5289			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	89.0803	6.6731	13.3493	0.0000	75.6958	102.4647	75.6958218	102.464718
CPI ratio	-54.7220	5.4228	-10.0911	0.0000	-65.5988	-43.8452	-65.598793	-43.845249

RESIDUAL OUTPUT

Observation	*Predicted mxn_eur*	*Residuals*	*andard Residuals*
1	25.51227212	-4.6413079	-3.04075376
2	25.24223937	-4.68388447	-3.06864781
3	25.30925946	-0.5901216	-0.38661828
4	25.93762021	0.348504663	0.228322897
5	25.78537582	-0.12709784	-0.08326817
6	25.48757867	-0.37224507	-0.24387643
7	25.31337842	0.41486945	0.2718018
8	25.00140262	1.265762717	0.829264687
9	24.52294013	1.005091386	0.658485814
10	24.10400889	0.938377295	0.614778065
11	24.19159408	0.000962461	0.000630557
12	24.08697318	0.200835746	0.131577577
13	23.62821392	0.65555533	0.4294872
14	23.21431096	1.372174825	0.8989806
15	23.06533105	1.630563909	1.068264258
16	22.91811019	1.057676173	0.692936748
17	22.99528788	1.254010564	0.821565261
18	22.73061671	1.403716509	0.91964514
19	22.37844611	1.224652248	0.802331155
20	22.65688621	0.965232249	0.632372093
21	22.09735989	1.486657836	0.973984166
22	21.81335917	1.896618235	1.2425698
23	21.28973111	2.535023869	1.660821374
24	21.17630479	2.471769216	1.619380075
25	20.9451787	2.262476615	1.482261988
26	20.89700309	2.320697644	1.520405506
27	21.18791755	1.448964669	0.94928948
28	21.11588437	0.58812663	0.385311274
29	21.45311887	-0.25795141	-0.16899692
30	21.38497254	-0.23987344	-0.15715313
31	21.07689843	-0.18485609	-0.1211085
32	20.91201802	-0.54003279	-0.35380259
33	20.09974983	-0.23397041	-0.15328576
34	20.4158559	-0.73242063	-0.47984551
35	20.2382813	-0.41065329	-0.26903958
36	19.91491371	0.858477227	0.56243152
37	19.7130886	0.731175659	0.479029873
38	20.03722435	-0.09164181	-0.06003915
39	20.45096481	-0.75368101	-0.49377426
40	20.87944633	-1.04391013	-0.68391792
41	20.99291678	-1.70498135	-1.11701885
42	21.04547155	-2.35603951	-1.54355974
43	20.75240438	-2.05723961	-1.34780093
44	21.05613486	-2.53454567	-1.66050808
45	20.39025139	-1.92097382	-1.25852637
46	20.2310678	-1.1525636	-0.75510226
47	19.67314085	-0.89152307	-0.58408151
48	19.2775917	-0.52140393	-0.34159788
49	18.53516248	0.094506694	0.061916079
50	19.07275986	-0.62357262	-0.40853372
51	19.02444844	-0.79593098	-0.52145434
52	19.23365071	-1.21661577	-0.79706606
53	19.38630177	-1.2217299	-0.80041658
54	19.175164	0.408202443	0.267433909
55	18.57243145	1.060085449	0.694515185

PROBABILITY OUTPUT

Percentile	*mxn_eur*
0.90909091	18.0170349
2.72727273	18.1645719
4.54545455	18.2285175
6.36363636	18.4491872
8.18181818	18.4692776
10	18.5215892
11.8181818	18.6296692
13.6363636	18.689432
15.4545455	18.6951648
17.2727273	18.7561878
19.0909091	18.7816178
20.9090909	19.0785042
22.7272727	19.2879354
24.5454545	19.5833664
26.3636364	19.6325169
28.1818182	19.6834353
30	19.6972838
31.8181818	19.827628
33.6363636	19.8355362
35.4545455	19.8657794
37.2727273	19.9455825
39.0909091	20.3719852
40.9090909	20.4442643
42.7272727	20.5583549
44.5454545	20.7733909
46.3636364	20.8709642
48.1818182	20.8920423
50	21.1450991
51.8181818	21.1951675
53.6363636	21.704011
55.4545455	22.6368822
57.2727273	23.2076553
59.0909091	23.2177007
60.9090909	23.5840177
62.7272727	23.6030984
64.5454545	23.6221185
66.3636364	23.648074
68.1818182	23.7099774
70	23.824755
71.8181818	23.9757864
73.6363636	24.1343332
75.4545455	24.1925565
77.2727273	24.2492984
79.0909091	24.2837692
80.9090909	24.2878089
82.7272727	24.5864858
84.5454545	24.695895
86.3636364	24.7191379
88.1818182	25.0423862
90	25.1153336
91.8181818	25.5280315
93.6363636	25.658278
95.4545455	25.7282479
97.2727273	26.2671653
99.0909091	26.2861249

Date	cpi_mex	cpi_fra	mxn_eur	CPI ratio
1/1/20	121.4392	104.54	20.87096422	1.16165296
2/1/20	121.9434	104.53	20.55835491	1.16658758
3/1/20	121.8853	104.59	24.71913787	1.16536285
4/1/20	120.6497	104.56	26.28612487	1.15388007
5/1/20	121.1141	104.71	25.65827798	1.15666221
6/1/20	121.7769	104.79	25.11533359	1.16210421
7/1/20	122.5766	105.19	25.72824787	1.16528757
8/1/20	123.0592	105.09	26.26716534	1.17098868
9/1/20	123.341	104.55	25.52803151	1.17973219
10/1/20	124.0939	104.51	25.04238619	1.18738781
11/1/20	124.1875	104.73	24.19255654	1.18578726
12/1/20	124.6609	104.96	24.28780893	1.18769912
1/1/21	125.7322	105.12	24.28376925	1.19608257
2/1/21	126.5273	105.12	24.58648578	1.20364631
3/1/21	127.5735	105.75	24.69589496	1.20636879
4/1/21	127.991	105.86	23.97578636	1.20905913
5/1/21	128.2523	106.2	24.24929844	1.20764878
6/1/21	128.9357	106.34	24.13433322	1.21248542
7/1/21	129.6932	106.4	23.60309836	1.21892105
8/1/21	129.9408	107.05	23.62211846	1.21383279
9/1/21	130.7416	106.81	23.58401772	1.22405767
10/1/21	131.8368	107.25	23.7099774	1.22924755
11/1/21	133.3462	107.64	23.82475498	1.23881643
12/1/21	133.8299	107.85	23.64807401	1.2408892
1/1/22	134.6216	108.12	23.20765531	1.24511284
2/1/22	135.7385	108.94	23.21770073	1.24599321
3/1/22	137.0824	110.49	22.63688222	1.24067698
4/1/22	137.824	110.97	21.704011	1.24199333
5/1/22	138.067	111.72	21.19516746	1.23583065
6/1/22	139.2329	112.55	21.1450991	1.23707597
7/1/22	140.2642	112.87	20.89204234	1.24270577
8/1/22	141.2396	113.38	20.37198523	1.24571882
9/1/22	142.1158	112.74	19.86577942	1.26056236
10/1/22	142.9201	113.9	19.68343527	1.25478578
11/1/22	143.7426	114.26	19.82762802	1.25803081
12/1/22	144.2914	114.16	20.77339094	1.26394008
1/1/23	145.2702	114.6	20.44426426	1.26762827
2/1/23	146.0802	115.78	19.94558254	1.26170496
3/1/23	146.4715	116.79	19.6972838	1.25414419
4/1/23	146.4419	117.5	19.8355362	1.24631404
5/1/23	146.1236	117.44	19.28793544	1.24424046
6/1/23	146.2719	117.65	18.68943204	1.24328007
7/1/23	146.9769	117.71	18.69516477	1.24863563
8/1/23	147.7904	118.89	18.52158919	1.2430852
9/1/23	148.4463	118.26	18.46927757	1.25525368
10/1/23	149.0042	118.43	19.0785042	1.25816263
11/1/23	149.958	118.23	18.78161777	1.26835828
12/1/23	151.0167	118.39	18.75618777	1.27558662
1/1/24	152.3651	118.19	18.62966918	1.2891539
2/1/24	152.5089	119.21	18.44918724	1.27932975
3/1/24	152.947	119.47	18.22851746	1.28021261
4/1/24	153.2561	120.07	18.01703493	1.27638961
5/1/24	152.9721	120.11	18.16457187	1.27360003
6/1/24	153.5505	120.2	19.58336644	1.2774584
7/1/24	155.1579	120.42	19.6325169	1.28847285